

Medical Report Intelligence Platform

Anti-Gravity Master Prompt Guide

Daily-Actionable Implementation Prompts for 30-Day Development

Version 1.0 | 2026

■ **How to Use This Guide:**

✓ Work on ONE phase per day | ✓ Use prompts as-is or adapt | ✓ Test each module independently | ✓ Follow AI safety principles always

PHASE 1: Foundation & Authentication (Days 1-3)

Day 1 | Module 1.1: User Authentication System

Anti-Gravity Prompt:

Build a lightweight user authentication module with email/password and social login (Google/Facebook) using Firebase/Auth0. The module should: (1) Handle user registration with email verification, (2) Support password reset flow, (3) Return JWT tokens for API authentication, (4) Store user metadata (name, email, phone) in PostgreSQL, (5) Enable logout and session management.

Acceptance Criteria: User can sign up → verify email → log in → receive token → call protected API.

Safety Check: Passwords never stored in plain text; HTTPS enforced.

Days 2-3 | Module 1.2: Database Schema & User Profile

Anti-Gravity Prompt:

Design PostgreSQL schema for users, medical reports, health records, and consultation history. Create tables: **users** (id, email, password_hash, phone, name, created_at), **medical_reports** (id, user_id, file_url, upload_date, report_type, confidence_score), **health_records** (id, user_id, report_id, diagnosis, medications, findings). Add indexes on user_id and report_type for fast queries. Set up soft deletes for GDPR compliance.

Acceptance Criteria: Schema created → test inserts/reads → queries run in <50ms.

Safety Check: All medical data encrypted at rest; audit logs for access.

PHASE 2: Data Intake Layer (Days 4-6)

Days 4-5 | Module 2.1: Medical Report Upload & Validation

Anti-Gravity Prompt:

Build a file upload module that: (1) Accepts PDF and JPG/PNG images up to 50MB, (2) Validates file type (magic bytes, not just extension), (3) Scans for malware using VirusTotal API, (4) Stores clean files in AWS S3 with encrypted object keys, (5) Generates unique file_id and returns upload metadata. Support resumable uploads for large files.

Acceptance Criteria: Upload 10MB PDF → virus scan passes → stored in S3 → file_id returned.

Safety Check: File size limits enforced; malware scanning mandatory before storage.

Day 6 | Module 2.2: OCR Engine (PaddleOCR Integration)

Anti-Gravity Prompt:

Integrate PaddleOCR to extract text from scanned medical reports and images. Workflow: (1) Receive S3 file_url, (2) Download from S3, (3) Run PaddleOCR with confidence filtering (>0.7), (4) Extract structured data: patient name, age, test dates, values, (5) Return OCR output + raw confidence scores. Cache OCR results for 7 days to save compute.

Acceptance Criteria: Scan blood report image → extract values with 92%+ accuracy → store in DB.

Safety Check: Confidence scores <0.7 flagged for manual review; never auto-diagnose from OCR alone.

PHASE 3: Intelligence & Explanation Layer (Days 7-12)

Days 7-8 | Module 3.1: Report Type Detection

Anti-Gravity Prompt:

Build a document classifier to identify report types: Blood Report, ECG, Chest X-ray, MRI, CT Scan, Ultrasound, etc. Use LayoutLMv3 or Donut model fine-tuned on 500+ sample reports. Process: (1) Extract text from OCR, (2) Pass to classifier, (3) Return top-3 predictions with confidence scores, (4) If confidence <85%, mark for manual review. Store classification result in medical_reports table.

Acceptance Criteria: Classify 50 diverse reports → 95%+ accuracy → sub-1s inference.

Safety Check: Low-confidence predictions always reviewed by humans; never auto-processed.

Days 9-11 | Module 3.2: Medical AI Explanation Engine

Anti-Gravity Prompt:

Build the core AI explanation module using LLM (GPT-5, Llama, or Gemini) with RAG on trusted medical literature. Workflow: (1) Parse OCR text + report type, (2) Retrieve 3-5 trusted medical documents via FAISS/Qdrant, (3) Prompt LLM: 'Explain these findings in simple language' + retrieved context, (4) Generate explanation with citations. **CRITICAL AI PRINCIPLES:** Never diagnose, prescribe, or recommend surgery. Always end with: 'Consult a qualified healthcare professional for medical advice.'

Acceptance Criteria: User uploads blood report → 2-3 paragraph simple explanation → cites sources.

Safety Check: All responses include disclaimer; diagnosis/prescription keywords blocked; manual QA on 10% of outputs.

Day 12 | Module 3.3: AI Confidence Meter Display

Anti-Gravity Prompt:

Display confidence scores for every AI output: OCR Quality (%), Report Type Detection (%), Explanation Confidence (%), Specialist Recommendation (%), Key Findings (%). Calculate composite score as weighted average: OCR(30%) + Classification(25%) + Explanation(25%) + Key Findings(20%). Show visual meter (0-100%) alongside each output. Add disclaimer: 'Confidence ≠ Medical Certainty. For guidance only.'

Acceptance Criteria: Every AI response shows 5 confidence scores; composite meter displays; disclaimer visible.

Safety Check: Never hide low-confidence metrics; always display uncertainty prominently.

PHASE 4: Recommendations & Guidance (Days 13-15)

Days 13-14 | Module 4.1: Specialist Recommendation

Anti-Gravity Prompt:

Build a specialist recommendation engine. Map report types → medical specialties: Blood Report → Hematologist/Internist, ECG → Cardiologist, Chest X-ray → Pulmonologist/Radiologist, MRI/CT → Radiologist, etc. Workflow: (1) Get report_type from classifier, (2) Look up recommended specialties (pre-built mapping), (3) Score each specialty on relevance (0-100), (4) Return top-3 specialists with reasoning.

Acceptance Criteria: Blood test uploaded → returns Hematologist + Internist with 95%+ confidence.

Safety Check: Recommendation is informational only; not a substitute for doctor consultation.

Day 15 | Module 4.2: Smart Doctor Question Generator

Anti-Gravity Prompt:

Generate smart questions users can ask their doctor during consultation. Input: OCR text + findings. Prompt LLM: 'List 5-7 simple, non-medical questions a patient should ask their doctor about these findings.' Questions should be: (1) Clear and patient-friendly, (2) Focused on impact (what does this mean for me?), (3) Action-oriented (what happens next?). Example: 'Will this affect my daily activities?' 'What follow-up tests are needed?'

Acceptance Criteria: Generate 6-7 relevant, non-technical questions; user finds them helpful.

Safety Check: Questions never imply diagnosis or treatment; always defer to doctor's expertise.

PHASE 5: Ecosystem Integration (Days 16-18)

Days 16-17 | Module 5.1: Nearby Hospital & Doctor Finder

Anti-Gravity Prompt:

Build location-based finder for hospitals and doctors. Use Google Places API + custom database of doctor profiles. Workflow: (1) Get user location (GPS or manual input), (2) Query 'hospitals within 5km' via Google Places API, (3) Filter by specialty (e.g., Cardiologists), (4) Sort by rating/distance, (5) Return top-10 with: name, address, phone, rating, distance, timings. Cache results for 24 hours.

Acceptance Criteria: User location 'Mumbai' → search 'Cardiologists' → returns 10 nearby doctors with ratings and distances.

Safety Check: Verify doctor credentials via medical board database; display license numbers.

Day 18 | Module 5.2: Appointment Integration & Redirect

Anti-Gravity Prompt:

Integrate appointment booking via doctor's online system (Practo, Zocdoc, or custom API). Workflow: (1) User clicks 'Book Appointment' on doctor profile, (2) Redirect to Practo/Zocdoc with pre-filled specialty, (3) Capture booking confirmation back in your app, (4) Send reminder SMS 24hrs before appointment. Fallback: Show doctor's phone number and email if online booking unavailable.

Acceptance Criteria: Click 'Book' → redirected to Practo → appointment booked → SMS reminder sent.

Safety Check: No payment processing in-app (use partner APIs); never store payment data.

PHASE 6: Advanced Features (Days 19-22)

Days 19-20 | Module 6.1: Medical Timeline View

Anti-Gravity Prompt:

Build an interactive timeline showing patient's health journey. Display: Report Date → Report Type → Key Findings → Specialist Seen → Status. Workflow: (1) Retrieve all reports for user sorted by date, (2) Extract key data points (dates, test names, results), (3) Render as vertical/horizontal timeline card UI, (4) Add filters (date range, report type), (5) Show trend (improving/worsening).

Acceptance Criteria: User sees 5 reports over 6 months visualized on timeline; can filter by date/type.

Safety Check: Display only user's own data; never share with third parties without explicit consent.

Day 21 | Module 6.2: Family Health Vault

Anti-Gravity Prompt:

Enable users to store and share family members' health records securely. Features: (1) Add family member (email invite), (2) Family member accepts and uploads own reports, (3) View family health overview (anonymous if preferred), (4) Emergency access: Set trusted contacts who can view records if unconscious, (5) Export family medical history as PDF.

Acceptance Criteria: User adds spouse → spouse uploads reports → user sees family dashboard → export PDF works.

Safety Check: Requires explicit consent for sharing; emergency access logged and auditable; encryption for all shared data.

Day 22 | Module 6.3: Report Comparison & Trend Analysis

Anti-Gravity Prompt:

Allow users to compare two medical reports side-by-side and see trends. Features: (1) Select 2 reports to compare, (2) Show findings in parallel columns, (3) Highlight changes (↑ improved, ↓ declined, → stable), (4) Extract key metrics (e.g., glucose: 120 → 105 mg/dL), (5) Generate trend sentence: 'Your glucose levels are improving over 3 months.'

Acceptance Criteria: Compare blood test from Jan vs Mar → see glucose drop highlighted → read summary.

Safety Check: Trend analysis is informational only; never prescribe based on trends; always recommend doctor consultation.

PHASE 7: AI Chat & Multilingual Access (Days 23-25)

Days 23-24 | Module 7.1: AI Medical Chat (RAG-Based)

Anti-Gravity Prompt:

Build a conversational AI chatbot using LLM + RAG over trusted medical literature. User can ask follow-up questions about their report. Workflow: (1) User: 'What does high cholesterol mean?' (2) Retrieve relevant documents from Qdrant on cholesterol, (3) Prompt LLM with context, (4) Generate conversational response, (5) Add source citations. Maintain conversation history (last 10 turns).

Acceptance Criteria: User uploads report → chats about findings → bot answers 5 questions with sources.

Safety Check: Every response: NEVER diagnose/prescribe → Always say 'Consult your doctor' → Cite sources → Block dangerous queries.

Day 25 | Module 7.2: Multilingual Translation Engine

Anti-Gravity Prompt:

Support 10+ languages (English, Hindi, Marathi, Tamil, Telugu, Kannada, Malayalam, Bengali, Gujarati, Punjabi). Workflow: (1) Detect user language from browser/settings, (2) Translate OCR text → local language, (3) Run AI explanation in local language, (4) Return all outputs (explanation, questions, specialist names) in user's language. Use Google Translate API + fine-tune on medical glossary.

Acceptance Criteria: Switch to Hindi → upload report → see OCR, explanation, questions all in Hindi.

Safety Check: Medical terms always verified with native speakers; maintain terminology accuracy across languages.

PHASE 8: User Engagement & Reminders (Day 26-27)

Days 26-27 | Module 8.1: Notification & Reminder System

Anti-Gravity Prompt:

Build notification engine for health reminders and alerts. Features: (1) Appointment reminder (SMS/email 24hrs before), (2) Follow-up reminder: 'Time for your follow-up test' (via doctor's recommendation), (3) Medication reminders (if linked to prescriptions), (4) Lab report available notification, (5) Health tips based on report findings. Use Firebase Cloud Messaging (FCM) for push, SNS for SMS. Users control notification frequency + opt-out anytime.

Acceptance Criteria: User gets appointment reminder 24hrs before → follows link → reads email → app shows notification.

Safety Check: Never share sensitive health info in notifications (subject line safe); allow opt-out anytime; GDPR compliant.

PHASE 9: Monitoring, Testing & Deployment (Days 28-30)

Days 28-29 | Module 9.1: Monitoring & Analytics

Anti-Gravity Prompt:

Set up Prometheus + Grafana for system monitoring and analytics dashboard. Track: (1) API response times (target <500ms), (2) OCR accuracy rate (%), (3) AI confidence distribution, (4) User upload volume, (5) Error rates by module, (6) Database query performance, (7) AI model inference time. Create alerts: OCR accuracy drops below 85% → alert team, API response >2s → log, errors >1% → page on-call.

Acceptance Criteria: Deploy Grafana → see live dashboards for all 7 metrics → alerts firing correctly.

Safety Check: PII never logged; audit all data access; rotate API keys monthly.

Day 30 | Module 9.2: Docker, Kubernetes & Production Deployment

Anti-Gravity Prompt:

Containerize entire platform with Docker + deploy on Kubernetes (K8s). Steps: (1) Build Docker images for FastAPI backend, React frontend, Redis cache, (2) Push to private ECR/Docker Hub, (3) Write K8s manifests: Deployment (replicas=3), Service, PVC for persistent storage, (4) Deploy to EKS/GKE, (5) Set up auto-scaling (CPU >70% → scale to 5 pods), (6) Configure SSL/TLS (Let's Encrypt), (7) Set up backup: daily snapshots of PostgreSQL + S3, (8) Run smoke tests post-deployment.

Acceptance Criteria: Deploy to production → all 15 modules running → handle 1000 concurrent users → <1% error rate.

Safety Check: Zero-downtime deployment with blue-green strategy; all secrets in AWS Secrets Manager; enable audit logging.

Cross-Cutting Concerns: Safety & Compliance (Throughout)

■ Data Security & Privacy

- All medical data encrypted at rest (AES-256) and in transit (TLS 1.3) • HIPAA/GDPR compliance: data anonymization, consent management, right to deletion • No third-party sharing without explicit consent • Audit logs for every data access event • Regular penetration testing (quarterly)

■ AI Safety & Ethics

- **Never diagnose, prescribe, or recommend surgery** • All AI outputs include disclaimer: 'Consult a healthcare professional' • Confidence scores displayed transparently • Manual review queue for low-confidence outputs (flagged automatically) • Monthly audits of AI model outputs for bias/accuracy • User feedback loop: collect incorrect explanations → retrain models

■ Testing Strategy

- Unit tests: 80%+ code coverage per module • Integration tests: simulate end-to-end workflows (upload → explain → recommend → book) • AI model tests: accuracy, precision, recall on test datasets • Security tests: OWASP Top 10, SQL injection, XSS, CSRF • Load tests: 1000 concurrent users, <500ms response time • User acceptance testing (UAT): real doctors + patients provide feedback

■ Documentation & Handoff

- API documentation (Swagger) with examples • Database schema documentation with ER diagrams • Deployment runbooks for production issues • On-call guide for common alerts and resolutions • User guides and FAQs for doctors + patients

Implementation Summary

Phase	Days	Modules	Key Deliverable
Foundation	1-3	Auth, DB Schema	Secure login + user profiles
Data Intake	4-6	Upload, OCR	Files extracted to text
Intelligence	7-12	Classification, Explanation	AI-powered report understanding
Recommendations	13-15	Specialist, Questions	Doctor-finding + consultation prep
Ecosystem	16-18	Hospitals, Appointments	Integrated healthcare access
Advanced	19-22	Timeline, Vault, Comparison	Health history management
AI & Accessibility	23-25	Chat, Multilingual	Conversational + global reach
Engagement	26-27	Notifications, Reminders	User retention & follow-ups
Deployment	28-30	Monitoring, K8s	Production-ready, scalable

Remember: Each day's work is independent and testable. Test before moving to the next phase. This ensures quality, catches issues early, and keeps the team unblocked. Use these prompts as conversation starters with your AI assistant—refine them based on your team's tech stack and constraints.